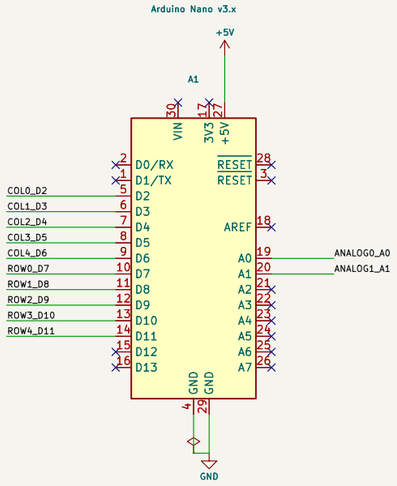
CONTROLLER PROTOTYPE | 25 buttons + 2 analog controls

Firmware-derived reference circuit. Assumed hardware: classic 5 V ATmega328P Nano, USB powered. See README.md.

USB / CONTROLLER



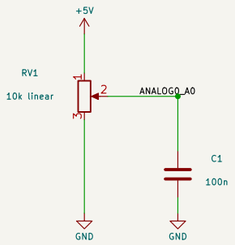
USB connector, USB-to-UART, reset, clock,

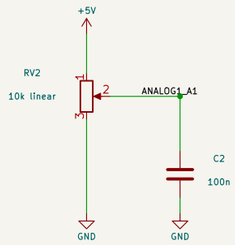
regulator and bypass capacitors are on A1.

USB carries power + 115200 baud serial data.

VIN, AREF and unused I/O: leave open.

ANALOG CONTROLS





RV1 -> analog channel 0; RV2 -> channel 1.

C1/C2: local wiper noise filtering.

PORTC = 0x00: A0/A1 pull-ups disabled at startup.

Pot dividers set ADC voltage without pull-up bias.

Keep both wipers connected; do not float inputs.

Build assumptions

A1: classic Nano v3.x, ATmega328P, 5 V / 16 MHz.

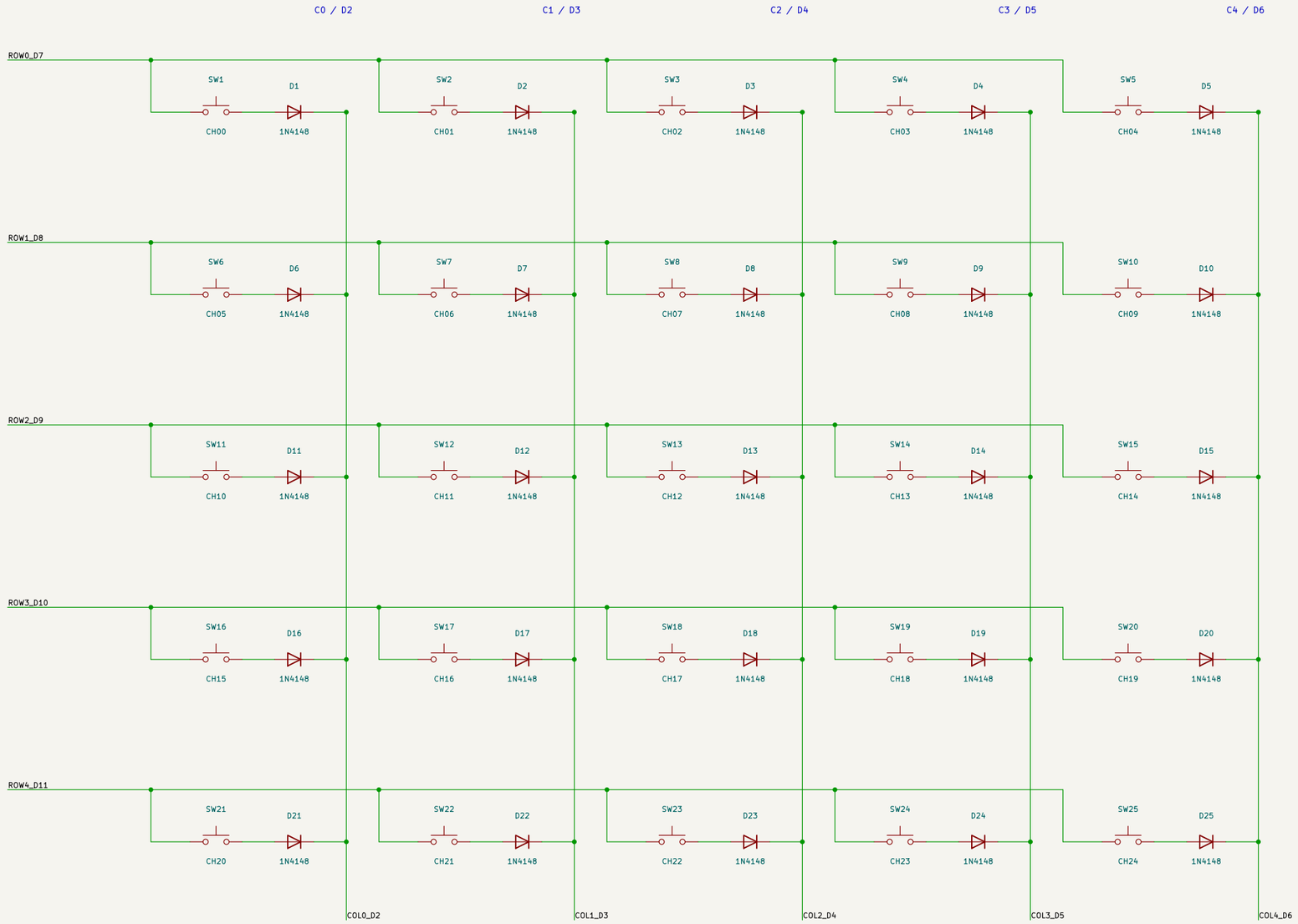
RV and switch mechanical parts are unspecified;

choose footprints after selecting actual controls.

Power the assembled module through USB only.

MATRIX: active LOW, internal pull-ups, no external row resistors

Each switch is normally open. Every diode: anode at switch; striped cathode at column.



CHANNEL ORDER (zero based): channel = row * 5 + column

SW1..SW5 = CH0..4, SW6..SW10 = CH5..9, ... SW21..SW25 = CH20..24.

One column is driven LOW; inactive columns and all rows use MCU pull-ups.

Diodes isolate simultaneous presses. Software debounce: 3 scans; update period: 10 ms.

Board and controls are implementation assumptions; see hardware/README.md
5 V ATmega328P Nano / USB powered / 25 digital + 2 analog
controller_prototype

Sheet: /
File: controller_prototype.kicad_sch

Title: Controller prototype – firmware-derived reference schematic

Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. 10.0.8		Id: 1/1